Tom S. **Bertalan**

APPLIED SCIENTIST IN DIFFERENTIABLE COMPUTING, PROBABILISTIC INFERENCE, AND NEUROSYMBOLIC PROGRAM SYNTHESIS; APPLIED TO COMPLEX SCIENTIFIC AND ENGINEERING SYSTEMS.

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EDUCATION

- **Princeton University** — ***2012 - 2018***
  Ph.D. & Masters, Chemical and Biological Engr.
- **The University of Alabama** — ***2008 - 2012***
  Bachelor of Science in Chemical and Biological Engr.; Minor in Math

PROFESSIONAL EXPERIENCE AND RESEARCH AREAS

Stealth Neurosymbolic AI Startup

FOUNDER / PRINCIPAL ENGINEER — *11/2025 - Present*

- Prototyping a system for automated scientific model discovery that combines symbolic representations of physical constraints with data-driven learning.
- Designing and implementing a neurosymbolic ML architecture integrating LLM-based semantic reasoning with differentiable physics simulations.
- Developing hybrid learning pipelines that incorporate physics-informed inductive biases alongside data-driven residual models to improve generalization under physical constraints.
- Building experimental pipelines in Python/PyTorch, including Neural ODEs, constrained optimization, and differentiable programming components.
- **Keywords:** Neurosymbolic AI, Physics-Informed ML (PIML/SciML), Neural ODEs, Differentiable Computing, Constrained Optimization, Generative Modeling, System Identification, Python, PyTorch

Amgen — *Cambridge, MA*

DATA SCIENTIST — *11/2024 - 11/2025*

- ***Mechanistic Modeling of Lyophilization and Filling for Biomanufacturing***
  - Designed and advocated a real-time lyophilization humidity digital twin using sequential Bayesian state estimation.
  - Conceived, implemented, and released company-wide a data-validated mechanistic simulation of lyophilization.
  - Calibrated model parameters using experimental data from seven production programs and designed a modular parameterization strategy to support pooled multi-product calibration workflows.
  - Contributed to a growing first-principles mechanistic model of peristaltic filling, used by a broad internal customer base.
  - Implemented data lake ingest pipelines for accelerometer and flow sensors into a new data product for the semantic data fabric.
  - Specified data schemas for a clean library interface, supporting connected and interoperable digital twin architectures across bioprocess operations, and rapid GUI generation on EC2-backed Plotly Dash platform.
  - Led agile sprint for demos for stakeholder requirement capture, and executed three iterative deliveries company-wide of containerized and versioned package releases.
- ***Hybrid Mechanistic+Black-box Modeling of CHO Cell Metabolism***
  - Evaluated vendor offerings for Gaussian process-based time series system identification tools.
  - Consulted on efficient adjoint gradient-based training of neural differential equations with embedded optimality constraints..
- ***Agile Development Automation Initiative***
  - Ingested handwritten notes using a multimodal large-language model on Databricks.
  - Accelerated sprint planning using the GitLab and Microsoft Graph APIs.

University of Massachusetts — *Lowell, MA*

RESEARCH SOFTWARE ENGINEER — *1/2024-11/2024*

- ***Neurosymbolic System Identification of Dynamical Systems***
  - Developed a neurosymbolic framework combining grammatical evolution for modular symbolic program synthesis with neural differential equations, enabling structured reasoning over scientific systems.
  - Ensured cross-domain modularity between robotics and bioprocessing through a domain-specific language design in ANTLR.
  - Prototyped an MVP for LLM-assisted symbolic code generation of differentiable physical simulators.
- ***Model Predictive Control of CHO Bioreactor***
  - Collected targeted experimental data from a medium-scale bioreactor and developed an online data-driven model-predictive controller with Aspen DMC3 to maximize economic factors, deploying to actual wet-lab hardware.
  - Supervised a rigorous implementation of a SBML-encoded bioreactor digital twin, including a sequential-Bayes soft-sensor for latent bioreactor glucose concentration.
  - Secured grant as PI to develop a central data lake for an NIH-seeded academic/industry consortium.

Johns Hopkins University *Baltimore, MD*

Postdoctoral Fellow *2/2020-1/2024*

- ***Neural Differential Equations and Time-Series***
  - Created a GUI for Bayesian experimental design; mentored team members on its use and maintenance.
  - Improved accuracy of backpropagation and convergence time of PyTorch neural stochastic differential equation training.
  - Advanced differentiable computing for learning continuous scientific laws from time-series data, bridging symbolic and statistical modeling through neural ODEs, PDEs, and SDEs; resulting in 9 publications and 10 conference presentations.
  - Cut RNN inference burn-in from 25 to 5 samples using manifold learning.
  - Led a team of biophysics and ML experts in creating a suite of simulation tools for CHO metabolism, including custom gradients for nets with constraints.
- ***Robotic Perception and Control***
  - Developed a variational autoencoder for end-to-end robotic localization.
  - Trained a U-net on both open and custom datasets for real-time (>10hz) on-board semantic segmentation of drivable space.

The Massachusetts Institute of Technology *Cambridge, MA*

Postdoctoral Associate *3/2018-2/2020*

- ***Nonlinear dynamics in neuroscience and bioprocessing***
  - Wrote object-oriented library for fine- and coarse-grained simulations of neuronal dynamics, probing bifurcation and resonance behavior of a mammalian circadian rhythm model.
  - Coauthored successful application for a $1.8M NIH-led industry-academic partnership grant for a new bioreactor simulation.
- ***Autonomous Vehicle Design and Pathfinding***
  - Developed a model AV with firmware-level speed sensing and control commanded by checksummed bus communications.
  - Simulated a comfort-optimized (jerk-minimizing) path planner with two-lane-ahead planning at 47 mph, and implemented a model-predictive path follower achieving 67 ms latency (IPOPT, CppAD) in C++ for real-time application.

Princeton University *Princeton, NJ*

NSF Research Assistant *9/2012-3/2018*

Built a rover for Bayesian particle-filter SLAM with LIDAR using Gazebo and ROS, and a custom library for visualizing OpenCV pipelines and dynamic pruning of computational graphs. Simulated thousands of neurons in Numpy and MATLAB, and social agents in OpenMP-accelerated C++. Advised undergrads on theses ranging from power-grid opt. with Gurobi to combinatorial basis extraction.

SKILLS AND INTERESTS

- Tensorflow/Pytorch
- LangChain and HuggingFace Transformers
- Databricks for data lakes and ML
- Python (15+ yrs), C++, Java
- Plotly, NumPy, SciPy, SymPy, Matplotlib, SKLearn
- Linux, shell scripting, and git
- OPC-UA, MODBUS/serial, and ROS
- Online optimization (IPOPT, CppAD, CasADi)
- Grammatical evolution for code generation
- Bayesian inference and probabilistic computing
- HPC with SLURM and OpenMP, and CUDA
- Differentiable physics for ML
- Mechanistic+data-driven grey-box modeling
- Retrieval-augmented generation (RAG)
- SQL(alchemy), HTML, CSS, JS, Flask, LAMP
- Classical and DL-based computer vision
- Trainee and peer mentoring
- Home automation with Arduino, Raspberry Pi, and 3D printing
- Solo and orchestral violin performance
- Windsurfing and small-boat sailing

SELECTED PUBLICATIONS AND PRESENTATIONS

Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets ***2024***
A. Zhu, B. Zhu, Tom Bertalan, Y. Tang, I. Kevrekidis *SIAMDS*

Symbolic regression and modular neural diff. eq. for bioproc. engr. and robotics ***2024***
Tom Bertalan, J. Lee, Z. Wang, S. Yoon, I. Kevrekidis *MLDS 4*

Certified Invertibility in Neural Networks via Mixed-Integer Programming ***2023***
T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab *L4DC 2023 — PMLR*

Learning effective SDEs from microscopic simulations: linking stoch. num. to deep learning ***2023***
F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, Tom Bertalan, S. Reich, I. Kevrekidis *Chaos*

Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators ***2022***
Y. Guo, F. Dietrich, Tom Bertalan, D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li *SIAM J. Sci. Comp.*

Local conformal autoencoder for standardized data coordinates ***2020***
E. Peterfreund, O. Lindenbaum, F. Dietrich, Tom Bertalan, M. Gavish, I. Kevrekidis, R. Coifman *PNAS*

PUBLICATIONS

Comparing Kinetic versus Stoichiometric Priorities in Hybrid Models of CHO Metabolism — ***2025 (under review)***
Tom Bertalan, P. Khare, M. Betenbaugh, I. Kevrekidis — *NPJ Sys. Bio. and Appl.*

Integ. of Bayesian Optim. and Solution Thermo. to Opt. Media Design for Mammalian Biomfg. — ***2025***
N. Ndahiro, E. Ma, Tom Bertalan, M. Donohue, Y. Kevrekidis, and M. Betenbaugh — *iScience*

Machine Learning Approaches to Problem Well-Posedness — ***2025***
Tom Bertalan, G. Kevrekidis, E. Rebrova, S. Mishra, I. Kevrekidis — *In preparation*

Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors — ***2024***
T. Cui, Tom Bertalan, N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis — *Comp. & Chem. Engr.*

Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets — ***2024***
A. Zhu, B. Zhu, Tom Bertalan, Y. Tang, I. Kevrekidis — *SIAMDS*

Transf. establishing equiv. across neural nets.: when have two nets. learned the same task? — ***2024***
Tom Bertalan, F. Dietrich, I. Kevrekidis — *Chaos*

Certified Invertibility in Neural Networks via Mixed-Integer Programming — ***2023***
T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab — *L4DC 2023 — PMLR*

Some of the variables, some of the parameters, some of the times, with some things known: Identification with partial information — ***2023***
S. Malani, Tom Bertalan, T. Cui, M. Betenbaugh, J. Avalos, I. Kevrekidis — *Comp. & Chem. Engr.*

Learning effective SDEs from microscopic simulations: linking stoch. num. to deep learning — ***2023***
F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, Tom Bertalan, S. Reich, I. Kevrekidis — *Chaos*

Learning emergent PDEs in a learned emergent space — ***2022***
F. Kemeth, Tom Bertalan, T. Thiem, S. Moon, C. Laing, I. Kevrekidis — *Nature Comm.*

Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators — ***2022***
Y. Guo, F. Dietrich, Tom Bertalan, D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li — *SIAM J. Sci. Comp.*

Initializing LSTM internal states via manifold learning — ***2021***
F. Kemeth, Tom Bertalan, N. Evangelou, T. Cui, S. Malani, I. Kevrekidis — *Chaos*

Dev. of closures for coarse-scale modeling of multiphase and free surface flows using ML — ***2021***
C. Linares, Tom Bertalan, E. Koronaki, Jicai Lu, G. Tryggvason, I. Kevrekidis — *APS Bulletin*

Global and local reduced models for interacting, heterogeneous agents — ***2021***
T. Thiem, F. Kemeth, Tom Bertalan, C. Liang, I. Kevrekidis — *Chaos*

Local conformal autoencoder for standardized data coordinates — ***2020***
E. Peterfreund, O. Lindenbaum, F. Dietrich, Tom Bertalan, M. Gavish, I. Kevrekidis, R. Coifman — *PNAS*

Emergent spaces for coupled oscillators — ***2020***
T. Thiem, M. Kooshkbaghi, Tom Bertalan, C. Liang, I. Kevrekidis — *Front. in Comp. Neuro.*

On Learning Hamiltonian Systems from Data — ***2019***
Tom Bertalan, F. Dietrich, I. Mezic, and I. Kevrekidis — *Chaos*

An Emergent Space for Distributed Data with Hidden Internal Order through Manifold Learning — ***2017***
Kemeth, Haugland, Dietrich, Tom Bertalan, Höhlein, Li, Bollt, Talmon, Krischer, & Kevrekidis — *IEEE Access*

Coarse-grained desc. of dynamics for nets with both intrinsic and structural heterogeneities — ***2017***
Tom Bertalan, Y. Wu, C. Laing, C. W. Gear, and I. Kevrekidis. — *Front. in Comp. Neuro.*

Dimension reduction in heterogeneous neural networks: Generalized Polynomial Chaos (gPC) and ANalysis-Of-VAriance (ANOVA) — ***2016***
M. Choi, Tom Bertalan, C. Laing, and I. Kevrekidis. — *Euro. Phys. J., Special Topics*

OpenMG: a new multigrid implementation in Python — ***2014***
Tom Bertalan, A. Islam, R. Sidje, and E. Carlson — *Num. Lin. Alg. with App.*

PRESENTATIONS

Integrated Modeling and Optim. of Secondary Drying for Pharma. Freeze-Drying Proc. — ***2025***
Tom Bertalan, M. Maritato, A. Ganguly, W. Al Bakri, R. Tukra, B. Lacetti — *AIChE 2025 (poster)*

Integrated Continuous USP Platform for Max. Productivity and Closed-Loop Controlled CQA — ***2025***
J. Lee, Tom Bertalan, Z. Wang, Y. Choi, S. Song, H. Yamanaka, A. Rajendran, J. Han, S. Yoon — *AIChE 2025*

Digital Twin-Enabled Control of Productivity and CQA in CHO Cell Cultures — ***2025***
Y. Choi, S. Yoon, Z. Wang, Tom Bertalan, J. Lee — *AIChE 2025 (poster)*

Beyond Trial-and-Error: An in silico Model for Accelerated Peristaltic Fill Recipe Development — ***2025***
Maritato, Ingram, Sao, Heymann, Bertalan, Banerjee, Içten-Gençer, Ganguly, Pearson, Schlegel — *AIChE 2025 (poster)*

Symbolic regression and modular neural diff. eq. for bioproc. engr. and robotics — ***2024***
Tom Bertalan, J. Lee, Z. Wang, S. Yoon, I. Kevrekidis — *MLDS 4*

Certified Invertibility in Neural Networks via Mixed-Integer Programming — ***2023***
T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab — *Learning for Dyn. Sys.*

Coarse-grained and emergent distributed-parameter systems from data — ***2021***
H. Arbabi, F. Kemeth, Tom Bertalan, I. Kevrekidis — *American Control Conf.*

Data-driven model reduction and discovery — ***2020***
T. Thiem, Tom Bertalan, F. Kemeth, Y. Psarellis, I. Kevrekidis — *AIChE*

Dynamical-systems-guided learning of PDEs from data — ***2020***
H. Arbabi, Tom Bertalan, A. Roberts, I. Kevrekidis — *AIChE*

On the data-driven discovery and calibration of closures — ***2020***
S. Lee, Y. Psarellis, C. Siettos, Tom Bertalan, D. Amchin, T. Bhattacharjee, S. Datta, I. Kevrekidis — *AIChE*

Connections between residual networks and explicit numerical integrators, and applications to identification of noninvertible dynamical systems — ***2020***
T. Cui, Tom Bertalan, Y. Psarellis, I. Kevrekidis — *AIChE*

Neural network approach to reduced order modeling of multiphase flows — ***2020***
C. Linares, Tom Bertalan, S. Lee, J. Lu, G. Tryggvason, I. Kevrekidis — *APS Div. of Fluid Dyn.*

PDE+PINN: Learning and Solving a PDE at the Same Time — ***2020***
Tom Bertalan, F. Kemeth, T. Cui, I. Kevrekidis — *AIChE*

Learning Partial Differential Equations from Discrete Space Time Data: Convolutional and Recurrent Networks, and Their Relations to Traditional Numerical Methods — ***2019***
Tom Bertalan, F. Dietrich, T. Thiem, R. Farber, I. Kevrekidis, A. Roberts — *AIChE*

Recurrent Neural Networks, Numerical Integrators and Nonlinear System Identification — ***2018***
Tom Bertalan, R. Farber, T. Thiem, F. Dietrich, I. Kevrekidis — *AIChE*

Coarse-Scale PDEs from Microscopic Observations Via Machine Learning — ***2019***
S. Lee, M. Kooshkbaghi, C. Siettos, Tom Bertalan, and I. Kevrekidis — *AIChE*

When Have Two Nets. Learned the Same Task? Data-Driven Transf. between System Reps. — ***2019***
Tom Bertalan, F. Dietrich, T. Thiem, I. Kevrekidis — *AIChE*

Coarse modeling of circadian rhythms in heterogeneous neural networks — ***2017; 2016***
Tom Bertalan, C. W. Gear, Y. G. Kevrekidis, M. Henson, E. Herzog, and C. Laing — *Dyn. Days 2017; AIChE*

Coarse-graining of heterogeneous neural dynamics — ***2015***
Tom Bertalan, M. Choi, C. Laing, I. Kevrekidis — *AIChE*

Heterogeneity and reduction for complex network dynamics — ***2014***
I. Kevrekidis, A. Holiday, Tom Bertalan, and C. Laing — *AIChE*

Coarse-graining Network Dynamics — ***2013***
A. Holiday and Tom Bertalan — *Network Front.*

nSpyres: an open-source, Python-based framework for sim. of flow through porous media — ***2012***
E. Carlson, A. Islam, F. Dumkwu, and Tom Bertalan — *Interpore 2012*

OpenMG: a new multigrid implementation in Python — ***2012***
Tom Bertalan, A. Islam, R. Sidje, and E. Carlson — *Proc. 11^{th} Python in Sci. Conf.*

ESIM: a framework for simulation of dominance hierarchy formation in small animal groups — ***2012***
Tom Bertalan and R. Earley — *UA Hon. Undergr. Res. Conf.*

An open-source computing cluster for virtual experiments with variable parameters — ***2011***
Tom Bertalan and E. Carlson — *UA Hon. Undergr. Res. Conf.*